OSCILLATIONS, WAVES AND OPTICS

PHYSICS FOR CHEMISTRY STUDENTS

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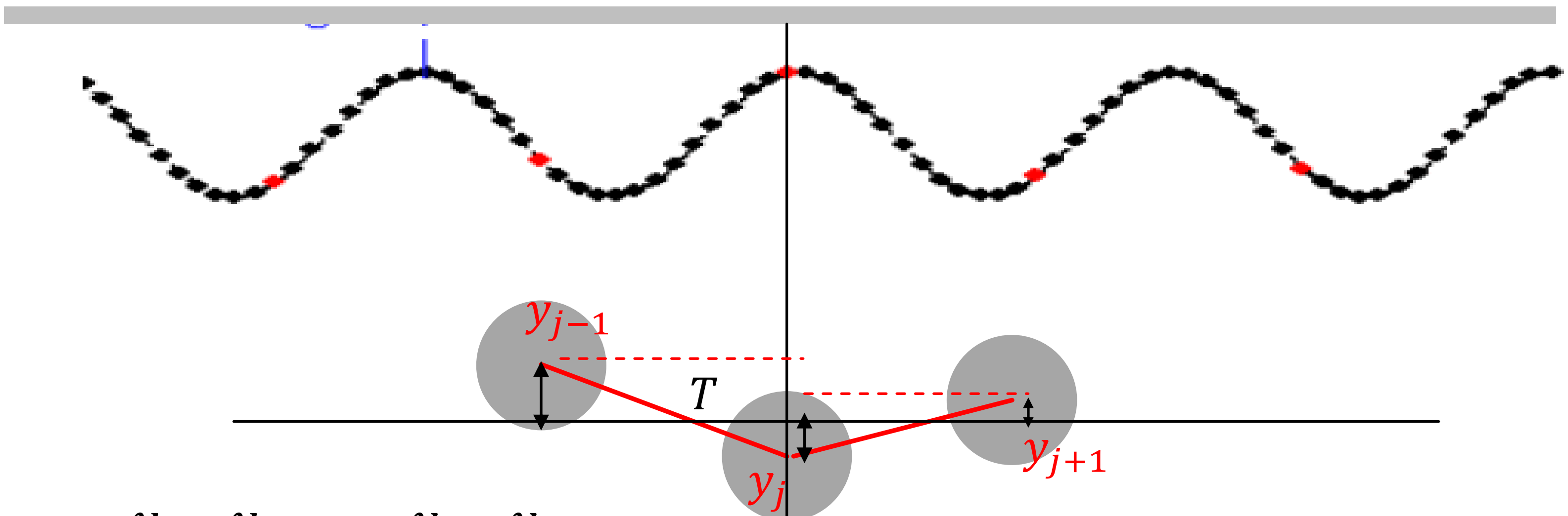
REVIEW



$$m\ddot{y}_j = -T\,\frac{y_j - y_{j-1}}{a} - T\,\frac{y_j - y_{j+1}}{a}$$

$$\ddot{\boldsymbol{Y}} = -\boldsymbol{W}\boldsymbol{Y} \qquad \boldsymbol{W} = \begin{pmatrix} \ddots & & 0 & & 0 \\ & 2\omega_0^2 & -\omega_0^2 & 0 & \\ 0 & -\omega_0^2 & 2\omega_0^2 & -\omega_0^2 & 0 \\ & 0 & -\omega_0^2 & 2\omega_0^2 & \\ 0 & & 0 & & \ddots \end{pmatrix}$$

Normal modes: $\boldsymbol{Y} = \boldsymbol{A}e^{i\omega_m t}$ $\quad m = 1, 2,3 \ldots$

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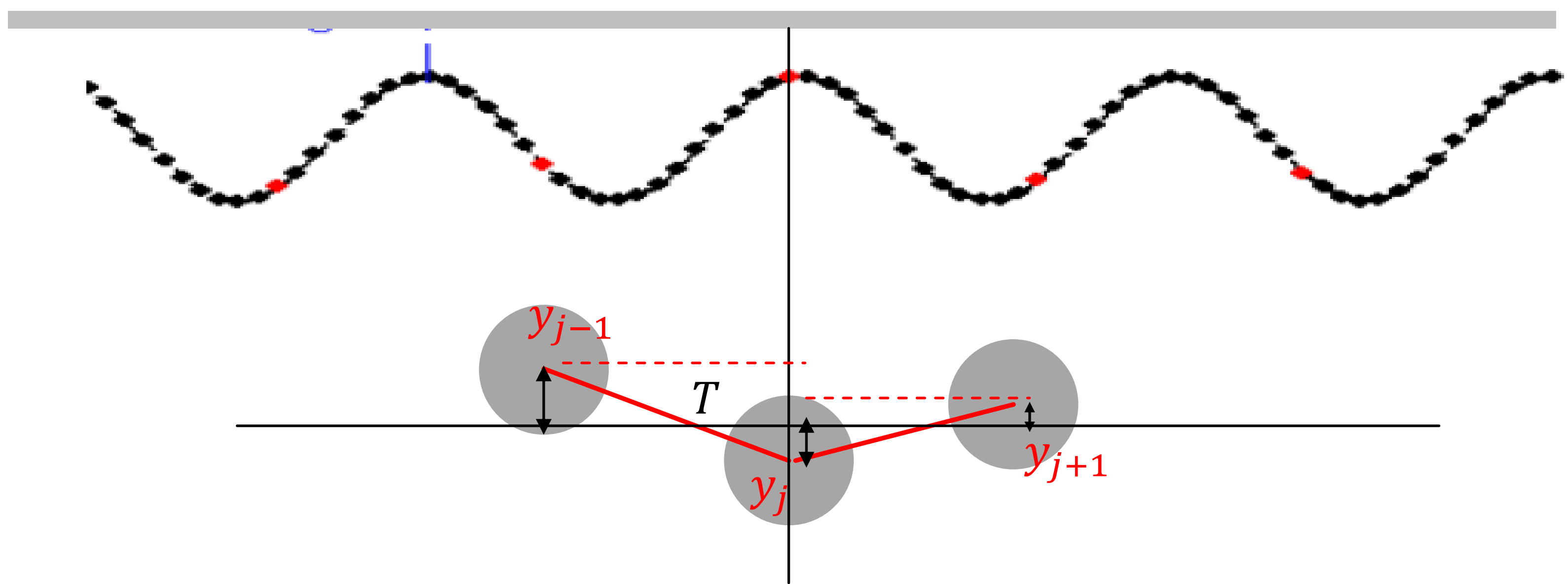

$$\ddot{y}_j = \left[-T \frac{\delta y}{\delta x}\bigg|_{x=x_j} + T \frac{\delta y}{\delta x}\bigg|_{x=x_{j+1}} \right] / (\rho \cdot \delta x) = \frac{T}{\rho}\left(y'\big|_{x+\delta x} - y'\big|_{x} \right) / \delta x = \frac{T}{\rho} y''\Big|_{x}$$

$$v_p^2 = \frac{T}{\rho} \qquad \boxed{\frac{\partial^2 y}{\partial t^2} = v_p^2 \frac{\partial^2 y}{\partial x^2}}$$

wave equation

REVIEW

$$\frac{\partial^2 u}{\partial t^2} = v_p^2 \frac{\partial^2 u}{\partial x^2}$$

$\xi = x + v_p t$

$\eta = x - v_p t$

$u(x,t) = u(\xi,\eta)$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial t} + \frac{\partial u}{\partial \eta}\frac{\partial \eta}{\partial t} = v_p\left(\frac{\partial u}{\partial \xi} - \frac{\partial u}{\partial \eta}\right)$$

$$\left.\begin{aligned}\frac{\partial^2 u}{\partial t^2} &= v_p^2\left(\frac{\partial^2 u}{\partial \xi^2} - 2\frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}\right)\\ \frac{\partial^2 u}{\partial x^2} &= \frac{\partial^2 u}{\partial \xi^2} + 2\frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}\end{aligned}\right\}$$

$$\frac{\partial^2 u}{\partial \xi^2} - 2\frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2} = \frac{\partial^2 u}{\partial \xi^2} + 2\frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}$$

$$\frac{\partial^2 u}{\partial \xi \partial \eta} = 0$$

General solution $u = f(\xi) + g(\eta)$

$$u(t,x) = f(x + v_p t) + g(x - v_p t)$$

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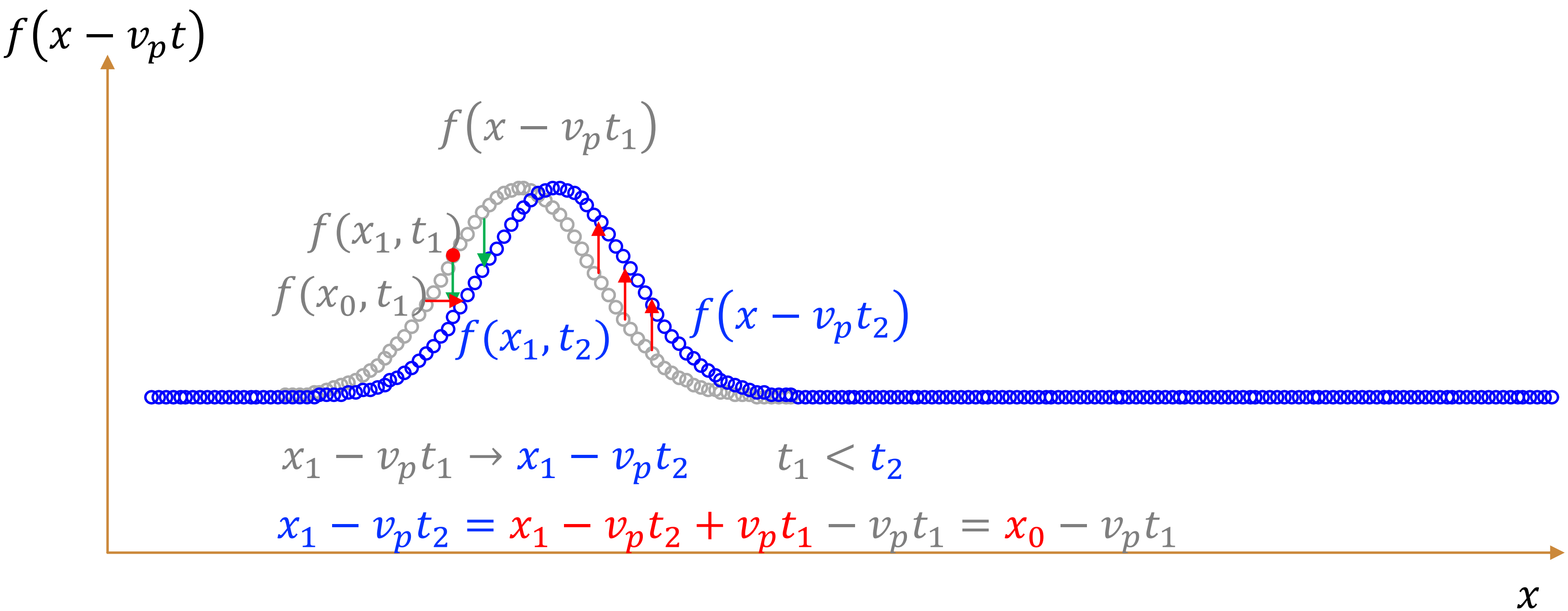

$f(x - v_p t)$ represents a wave travels forward

$g(x + v_p t)$ represents a wave travels backward

REVIEW

1D Free Waves: $$\frac{\partial^2 u}{\partial t^2} = v_p^2 \frac{\partial^2 u}{\partial x^2} (-\infty < x < \infty, t > 0)$$

initial condition: $$u(0,x) = \varphi(x); \quad \dot{u}(0,x) = \phi(x)$$

general solution: $$u(t,x) = f(x + v_p t) + g(x - v_p t)$$

$$u(0,x) = f(x) + g(x) = \varphi(x)$$

$$\dot{u}(0,x) = v_p f' - v_p g' = \phi(x) \qquad f(x) - g(x) = \frac{1}{v_p}\int_0^x \phi(\xi) d(\xi) + c$$

$$u(t,x) = f(x + v_p t) + g(x - v_p t) = \frac{\varphi(x + v_p t) + \varphi(x - v_p t)}{2} + \frac{1}{2v_p}\int_{x - v_p t}^{x + v_p t} \phi(\xi)\, d\xi$$

d'Alembert's formula

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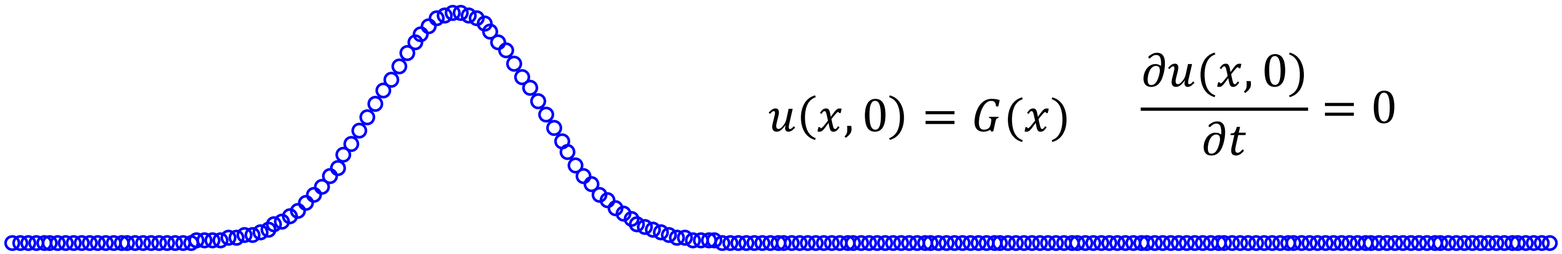
Initial condition:



$$u(x,0) = G(x) \qquad \frac{\partial u(x,0)}{\partial t} = 0$$

$$G(x) = ae^{-\frac{(x-x_0)^2}{2c^2}}$$

$u(x, t = T)\ ???$

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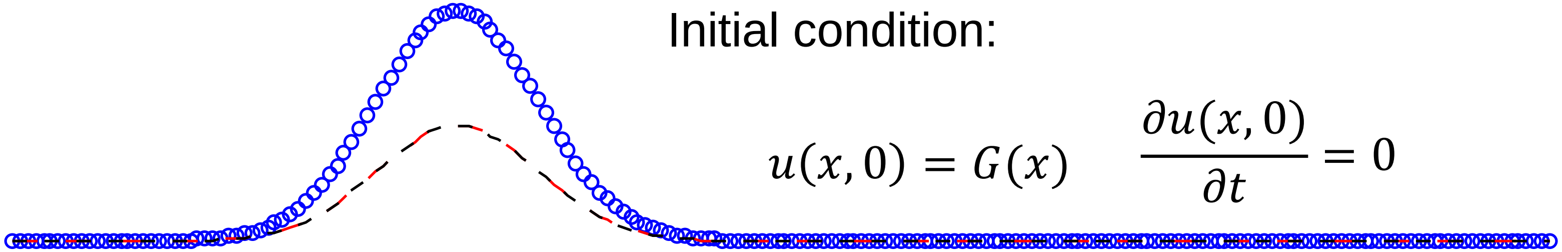

Initial condition:

$$u(x,0) = G(x) \qquad \frac{\partial u(x,0)}{\partial t} = 0$$

$$v_p = \sqrt{\frac{T}{\rho}}$$

$$u(x,t) = f\left(x - v_p t\right) + g\left(x + v_p t\right)$$

$$u(x,0) = f(x) + g(x) = G(x) \qquad \left.\frac{\partial u(x,t)}{\partial t}\right|_{t=0} = 0$$

$$u(x,t) = \frac{G\left(x + v_p t\right) + G(x - v_p t)}{2} + \frac{1}{2v_p}\int_{x-v_p t}^{x+v_p t} 0 \, d\xi = \frac{G\left(x + v_p t\right) + G(x - v_p t)}{2}$$

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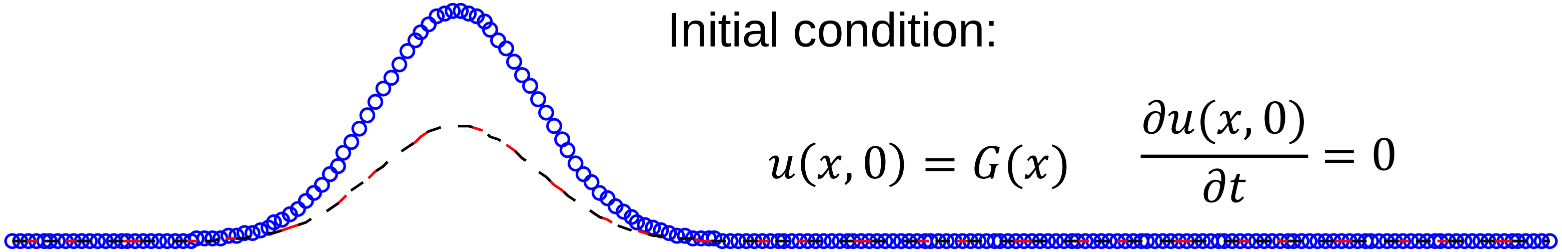
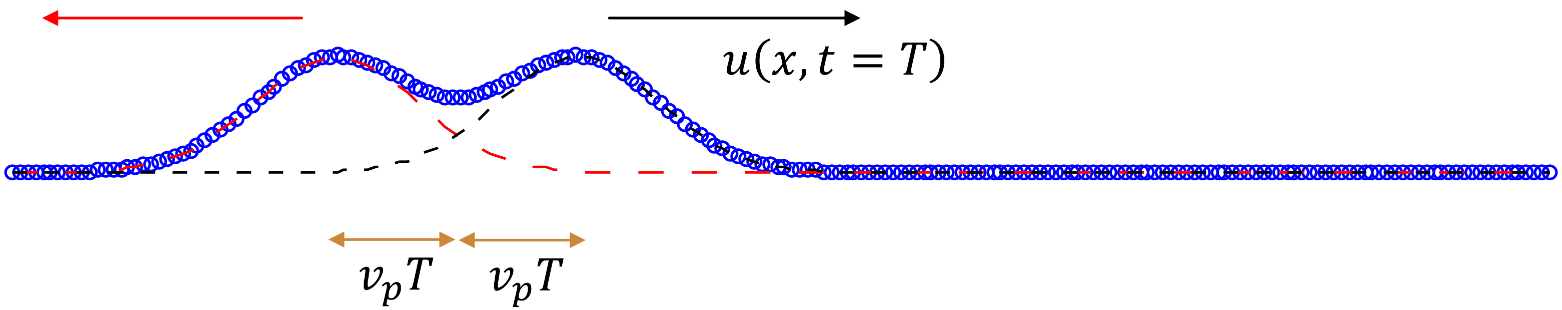


Initial condition:

$$u(x,0) = G(x) \qquad \frac{\partial u(x,0)}{\partial t} = 0$$

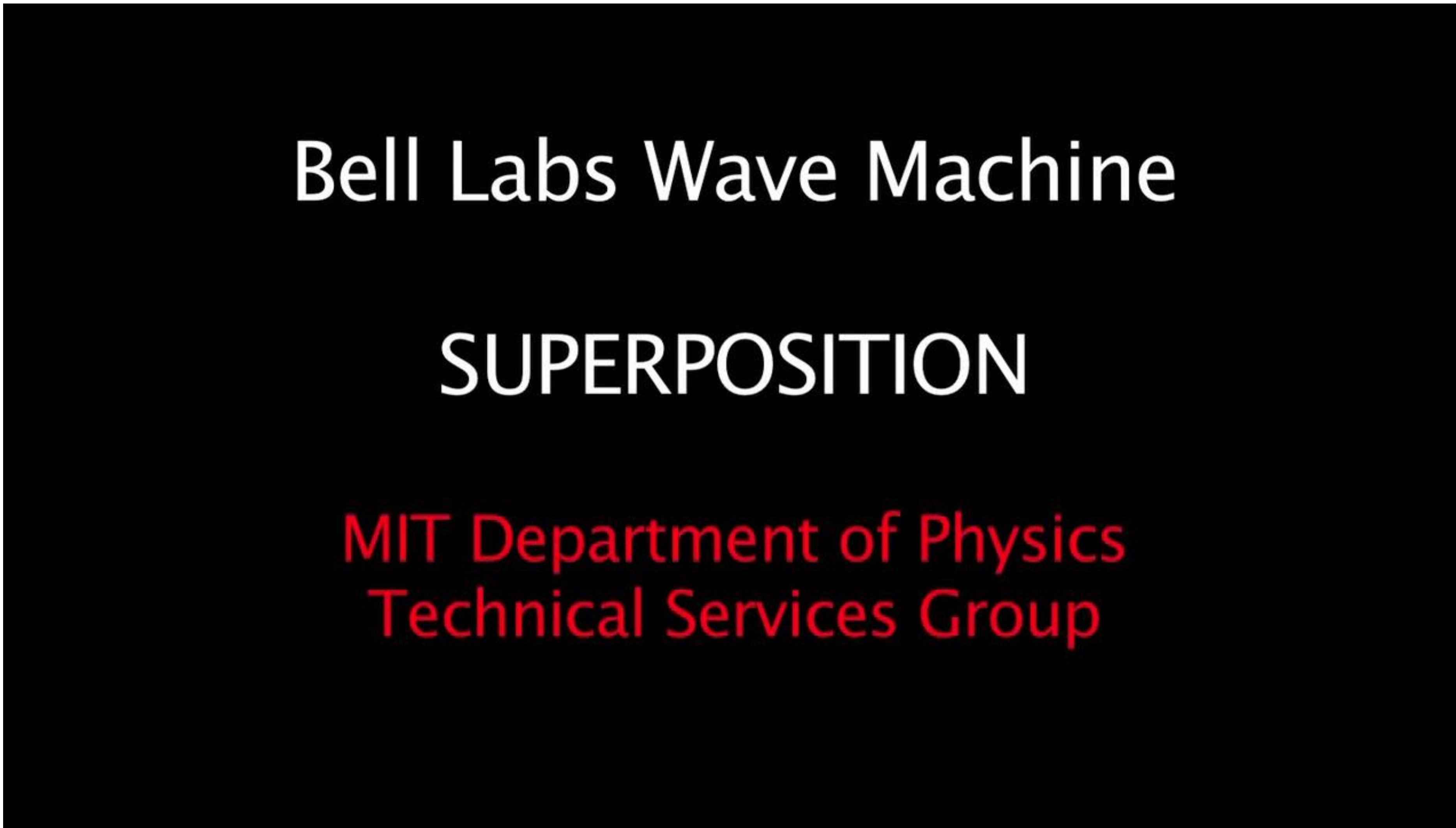
$$u(x,t) = \frac{1}{2}G\left(x - v_p t\right) + \frac{1}{2}G\left(x + v_p t\right) \qquad v_p = \sqrt{\frac{T}{\rho}}$$

REVIEW

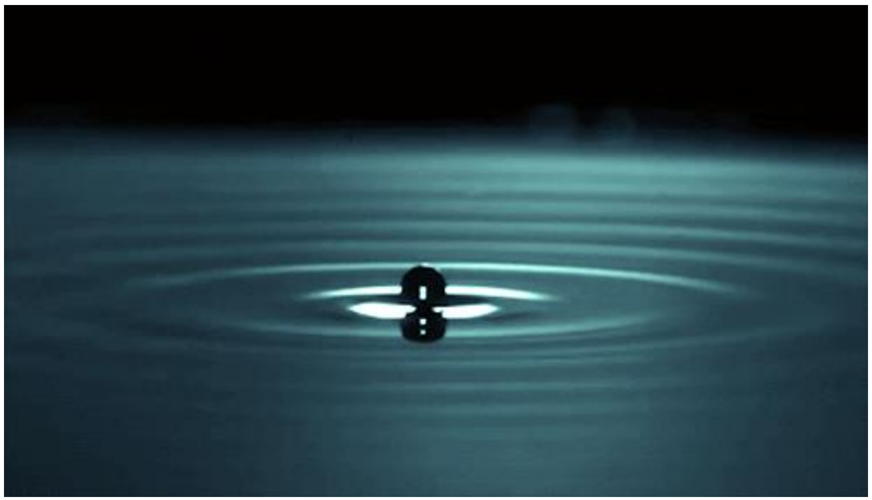


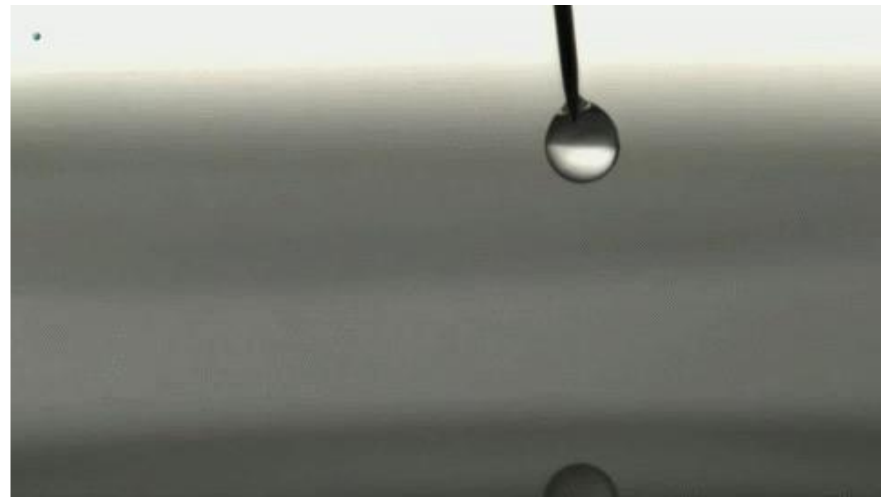
$$u(x,t) = \frac{1}{2} G\left(x - v_p t\right) + \frac{1}{2} G\left(x + v_p t\right) \qquad v_p = \sqrt{\frac{T}{\rho}}$$

REVIEW


Bell Labs Wave Machine
SUPERPOSITION
MIT Department of Physics
Technical Services Group

REVIEW





REVIEW

$$\frac{\partial^2 u}{\partial t^2} = v_p^2 \frac{\partial^2 u}{\partial x^2}$$

Separation of variables

*m*th normal mode for continuous wave equation

$$u_m(x,t) = c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$

Dispersion relation: $v_p = \frac{\omega}{k}$

Normal Modes !

General solution

$$u(x,t) = \sum_m c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$

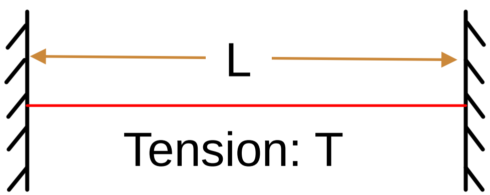
$c_m,\ k_m,\ \varphi\ and\ \phi$ are unknown

REVIEW

*m*th normal mode for continuous wave equation

$$u_m(x,t) = c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$

In specific cases: Two fix ends



$$u_m(0,t) = c_m \sin(\varphi)\sin(\omega_m t + \phi) = 0$$

$$\varphi = n\pi$$

$$u_m(L,t) = c_m \sin(k_m L + \varphi)\sin(\omega_m t + \phi) = 0$$

Boundary conditions:

$$u(0,t) = 0$$

$$u(L,t) = 0$$

$$k_m L + \varphi = n'\pi$$

$$k_m = m\pi/L$$

$$\vdots$$

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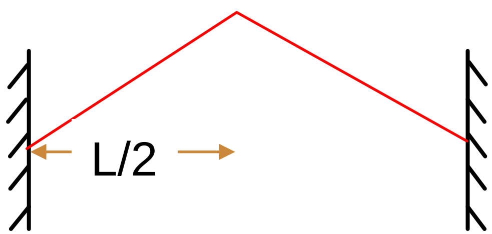

Initial condition t=0:

General solution

$$u(x,t) = \sum_m c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$

Boundary conditions: $k_m = \frac{m\pi}{L}$

Initial condition

$$\dot{u}(x,0) = \sum_m c_m \omega_m \sin(k_m x + \varphi)\cos(\phi) = 0$$

$$\phi = \frac{n\pi}{2}, n = 1,3,5 \ldots$$

$$u(x,0) = \sum_m c_m \sin(k_m x + \varphi) \cdot 1 = \sum_{m=0,1,2\ldots} a_m \cos(k_m x) + b_m \sin(k_m x)$$

Fourier series

$$c_m = \sqrt{{a_m}^2 + {b_m}^2}$$

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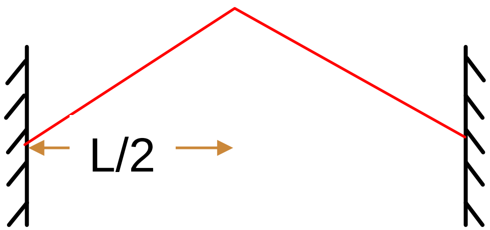

even function

Initial condition t=0:

$$u(x,0) = \sum_m c_m \sin(k_m x + \varphi) \cdot 1 = \sum_{m=0,1,2\ldots} a_m \cos(k_m x) + \cancel{b_m \sin(k_m x)}$$

Fourier series

$$c_m = \sqrt{{a_m}^2 + {b_m}^2}$$

$$u(x,0) = \sum_{m=0,1,2\ldots} c_m \cos(k_m x)$$

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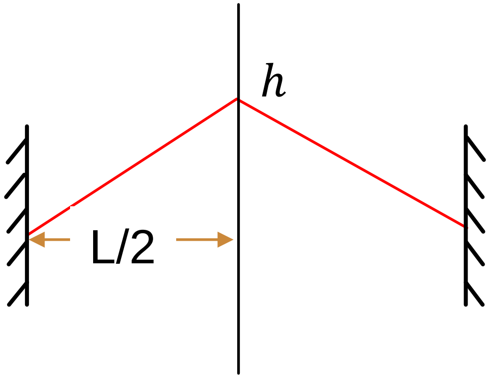


Initial conditions:

$$u(x,0) = -\frac{2h}{L}x + h; 0 < x < \frac{L}{2}$$

$$u(x,0) = \frac{2h}{L}x + h; -\frac{L}{2} < x < 0$$

$$= \frac{4}{L}\int_{-L/2}^{0} \frac{2h}{L} x \cos\left(\frac{2\pi}{L}mx\right) dx + \frac{4}{L}\int_{-L/2}^{0} h \cos\left(\frac{2\pi}{L}mx\right) dx$$

$$c_m = -2h\left(\frac{1}{m\pi}\right)^2 \cos(m\pi) + 2h\left(\frac{1}{m\pi}\right)^2$$

When m is even, $c_m = 0$ When m is odd, $c_m = 4h\left(\frac{1}{m\pi}\right)^2$

3.6 TRAVELING WAVE

$$\frac{\partial^2 u}{\partial t^2} = v_p^2 \frac{\partial^2 u}{\partial x^2}$$

General solution of the wave equation

$$u(x,t) = g(x + v_p t) + f(x - v_p t)$$

$$u_m(x,t) = \frac{c_m}{2}\cos\left[k_m\left(x - v_p t + \frac{\varphi'}{k_m}\right)\right] - \frac{c_m}{2}\cos\left[k_m\left(x + v_p t + \frac{\phi'}{k_m}\right)\right]$$

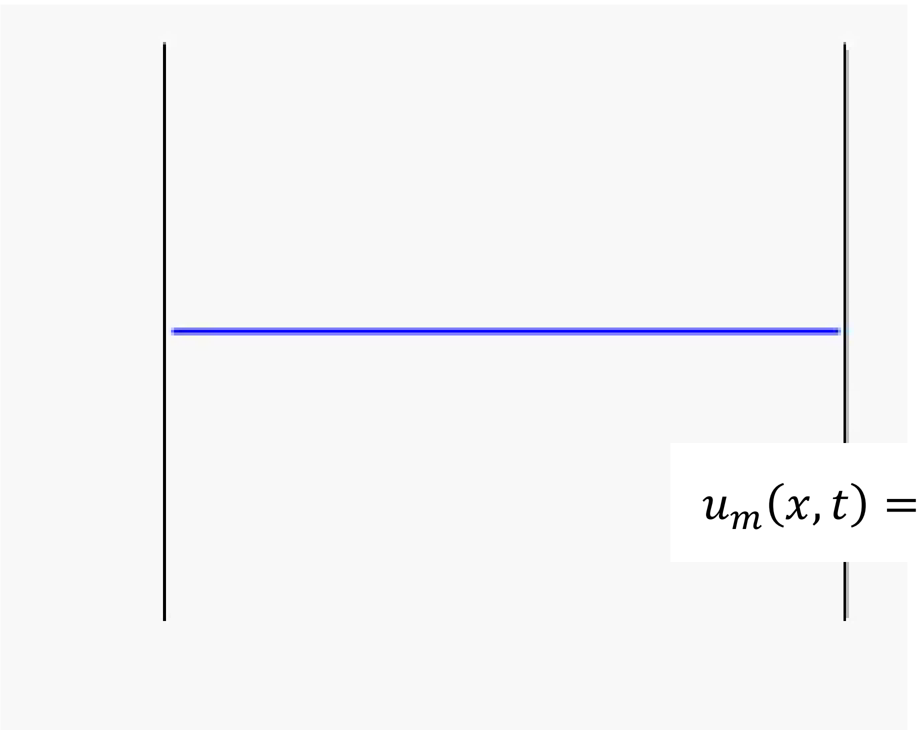
General solution

$$u_m(x,t) = c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$

$$\frac{\omega_m}{k_m} = v_p$$

$$u_m(x,t) = \frac{c_m}{2}\cos(k_m x - \omega_m t + \varphi') - \frac{c_m}{2}\cos(k_m x + \omega_m t + \phi')$$

3.6 TRAVELING WAVE



$$u_m(x,t) = c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$

Standing wave:

$$u_m(x,t) = c_m \sin(k_m x)\sin(\omega_m t)$$

$$u_m(x,t) = \frac{c_m}{2}\cos(k_m x - \omega_m t) - \frac{c_m}{2}\cos(k_m x + \omega_m t)$$

$$u_m(x,t) = F\left(x - \frac{\omega_m}{k_m}t\right) - F\left(x + \frac{\omega_m}{k_m}t\right)$$

A standing wave is a combination of traveling waves going in opposite directions. Likewise, a traveling wave is a combination of standing waves.

3.6 TRAVELING WAVE

$$F\left(x - v_p t\right) = Aexp(-\frac{\left(x - v_p t\right)^2}{2\sigma^2})$$

time = 0

3.6 TRAVELING WAVE

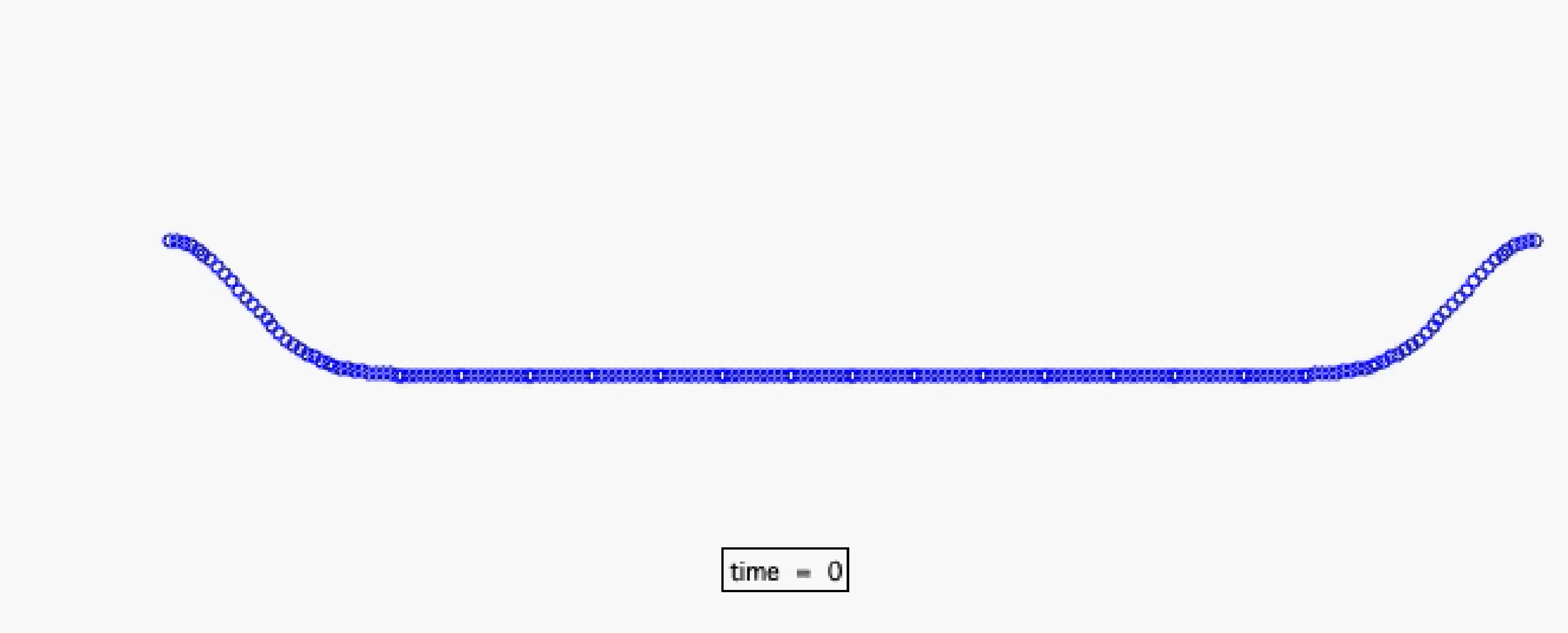
$$u(x,t) = \frac{5}{2 + e^{(3x+6t)}}$$

What the velocity of this traveling wave?

Method 1. $$\frac{\partial^2 u}{\partial t^2} = v_p^2 \frac{\partial^2 u}{\partial x^2}$$

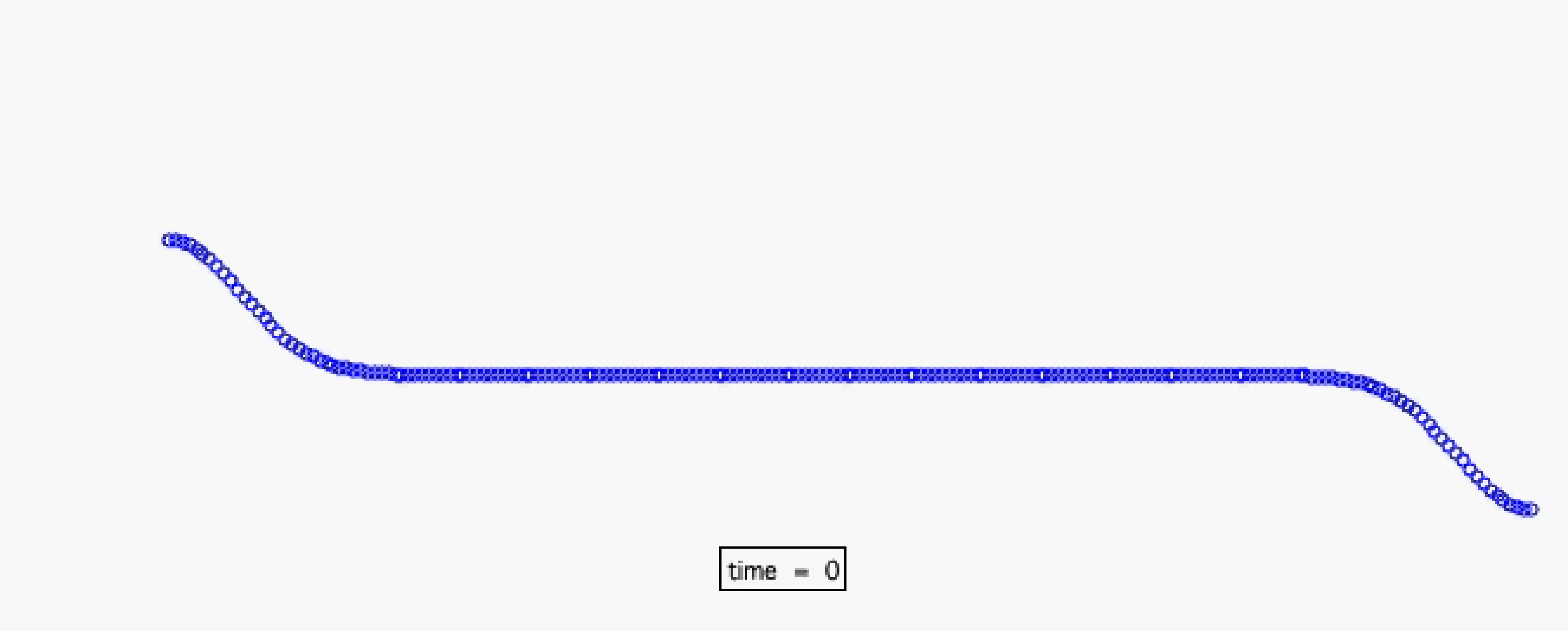
Method 2. $$u(x,t) = \frac{5}{2 + e^{3(x+2t)}}$$

3.6 TRAVELING WAVE



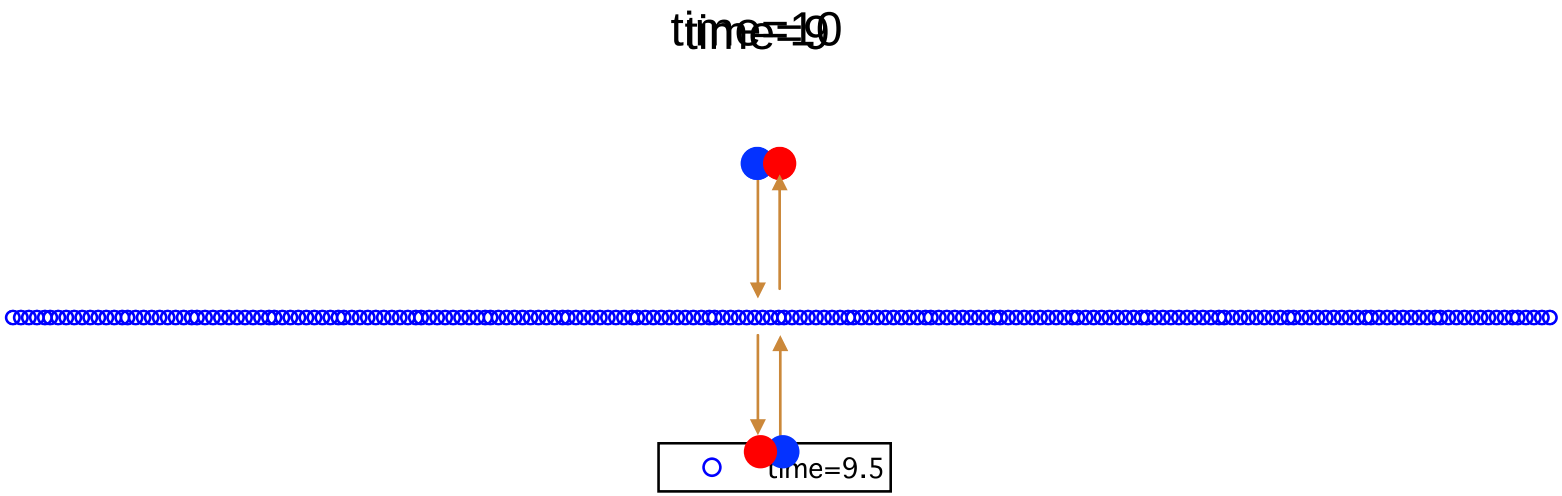
$$u(x,t) = f\left(x - v_p t\right) + g\left(x + v_p t\right)$$

3.6 TRAVELING WAVE



$$u(x,t) = f\left(x - v_p t\right) - g\left(x + v_p t\right)$$

3.7 ENERGY STORED IN THE STRING

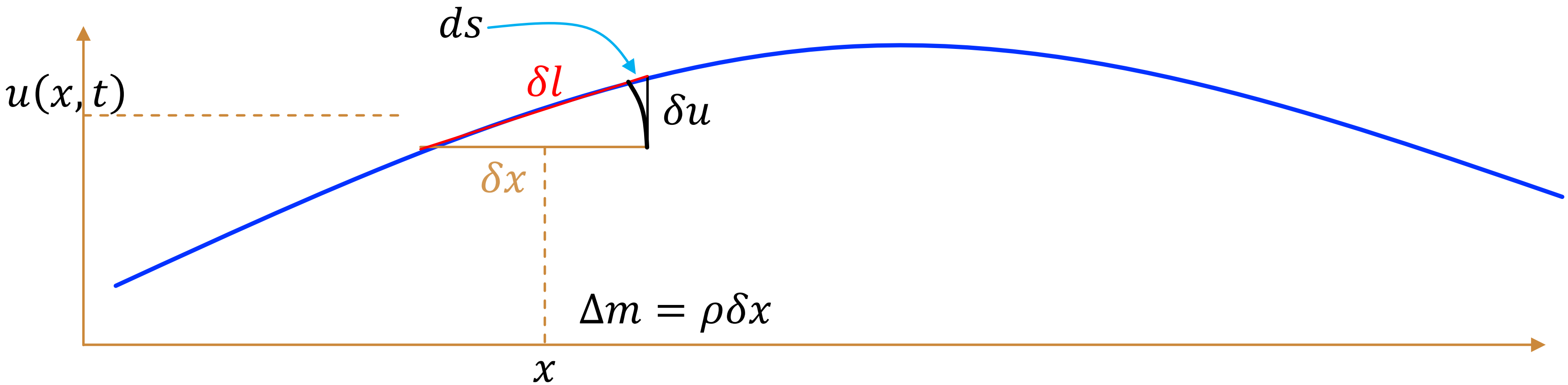


$$u(x,t) = f\left(x - v_p t\right) + g\left(x + v_p t\right)$$

Energy annihilation???

Potential energy→ Kinetic energy

3.7 ENERGY OF WAVE



Potential energy density:

$$ds = \delta l - \delta x = \sqrt{(\delta u)^2 + (\delta x)^2} - \delta x = \delta x\left(\sqrt{\left(\frac{\delta u}{\delta x}\right)^2 + 1} - 1\right) = \frac{1}{2}\delta x\left(\frac{\delta u}{\delta x}\right)^2$$

$$\delta E_V = T ds = \frac{1}{2}T\delta x\left(\frac{\delta u}{\delta x}\right)^2$$

$$\epsilon_V(x) = \frac{\delta E_V}{\delta x} = \frac{1}{2}T\left(\frac{\partial u}{\partial x}\right)^2$$

Kinetic energy density:

$$dE_K = \frac{1}{2}\rho\delta x \cdot \dot{u}^2(x,t)$$

$$\epsilon_K(x) = \frac{\delta E_K}{\delta x} = \frac{1}{2}\rho\left(\frac{\partial u}{\partial t}\right)^2$$

3.7 ENERGY OF WAVE

Potential Energy Density:

$$\epsilon_V = \frac{1}{2}T\left(\frac{\partial u}{\partial x}\right)^2$$

Kinetic Energy Density:

$$\epsilon_K = \frac{1}{2}\rho\left(\frac{\partial u}{\partial t}\right)^2$$

A forward-propagating wave: $u(x,t) = f(x - v_p t)$

$$\frac{\partial u}{\partial x} = -\frac{1}{v_p}\frac{\partial u}{\partial t}$$

$$\begin{cases} \dfrac{\partial u}{\partial t} = \dfrac{\partial u}{\partial \xi}\dfrac{\partial \xi}{\partial t} = -v_p\dfrac{\partial u}{\partial \xi} \\ \dfrac{\partial u}{\partial x} = \dfrac{\partial u}{\partial \xi}\dfrac{\partial \xi}{\partial x} = \dfrac{\partial u}{\partial \xi} \end{cases}$$

$$\epsilon_V = \frac{1}{2}T\left(\frac{\partial u}{\partial x}\right)^2 = \frac{1}{2}T\frac{1}{v_p^2}\left(\frac{\partial u}{\partial t}\right)^2 = \frac{1}{2}\rho\left(\frac{\partial u}{\partial t}\right)^2 = \epsilon_K$$

Total energy density is equally divided.

3.7 ENERGY OF WAVE

Potential Energy Density:

$$\epsilon_V = \frac{1}{2} T \left(\frac{\partial u}{\partial x}\right)^2$$

Kinetic Energy Density:

$$\epsilon_K = \frac{1}{2} \rho \left(\frac{\partial u}{\partial t}\right)^2$$

A forward-propagating wave: $u(x,t) = A\cos(kx - \omega t)$

$$\epsilon_V = \epsilon_K = \frac{1}{2}\rho\omega^2 A^2 \sin^2(kx - \omega t)$$ periodic function

Average Potential Energy Density and Kinetic Energy Density:

$$\overline{\epsilon_V} = \overline{\epsilon_K} = \frac{1}{\lambda}\int_0^\lambda \frac{1}{2}\rho\omega^2 A^2 \sin^2(kx - \omega t)dx = \frac{1}{4}\rho\omega^2 A^2$$

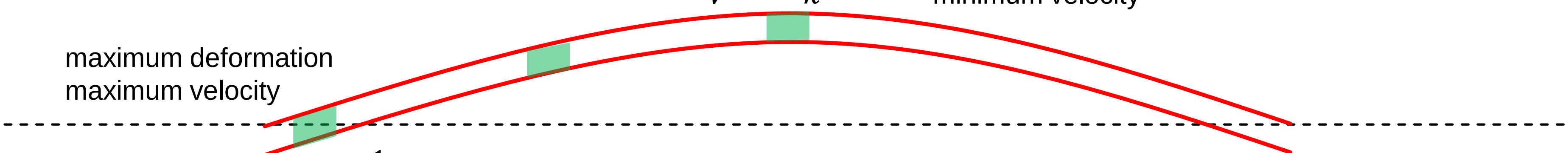
Average total energy density is

$$\bar{\epsilon} = \frac{1}{2}\rho\omega^2 A^2$$

3.7 ENERGY OF WAVE

$\epsilon_V = \epsilon_k = 0$ minimum deformation minimum velocity

maximum deformation
maximum velocity



$$\epsilon_V = \epsilon_k = \frac{1}{2}\rho\omega^2 A^2$$

wave

different

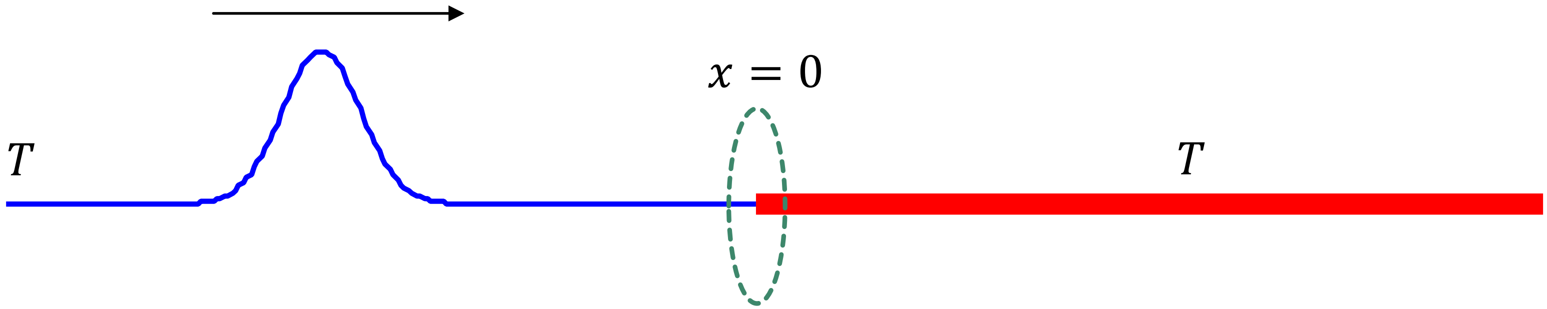
Harmonic oscillator

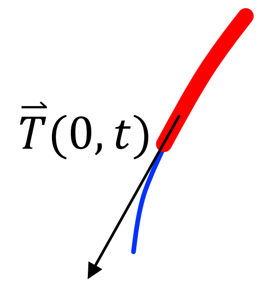
Kinetic Energy: $E_k = \frac{1}{2}m\dot{x}^2 = \frac{1}{2}m\omega^2 A^2 sin^2(\omega t + \varphi)$

Potential Energy: $E_k = \frac{1}{2}kx^2 = \frac{1}{2}kA^2 cos^2(\omega t + \varphi)$

3.8 REFLECTION AND TRANSMISSION

$$u(x,t) = F\left(x - v_p t\right)$$





1. $u(0+,t) = u(0-,t)$

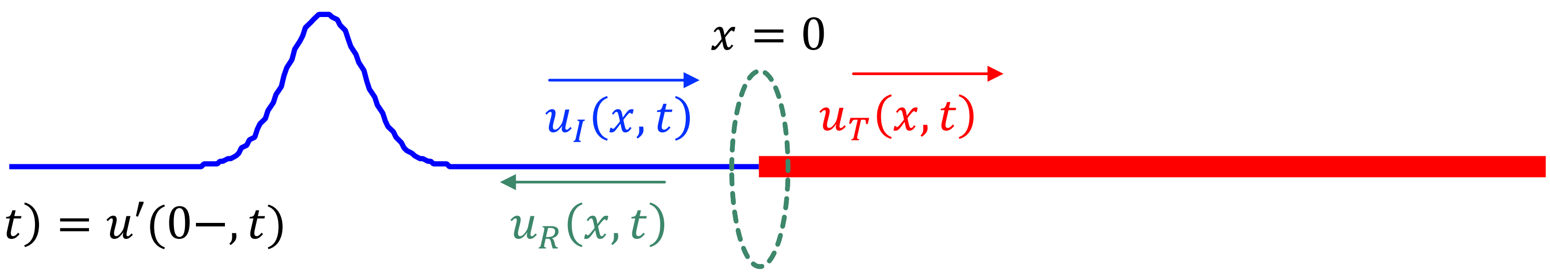
2. $u'(0+,t) = u'(0-,t)$ $\quad \Leftarrow \quad$ $\vec{T}(0+,t) = \vec{T}(0-,t)$

Continuity of the matter and force.

$$\ddot{u} = \delta T/\delta m = \delta T/\rho\delta x$$

$$\delta x \to 0 \leftrightarrow \delta T \to 0$$

3.8 REFLECTION AND TRANSMISSION



$$u'(0+,t) = u'(0-,t)$$

$$u_I{}'(0+,t) + u_R{}'(0+,t) = u_T{}'(0-,t)$$

$$\left.\frac{du_I}{d\xi}\frac{\partial \xi}{\partial x} + \frac{du_R}{d\zeta}\frac{\partial \zeta}{\partial x} = \frac{du_T}{d\eta}\frac{\partial \eta}{\partial x}\right|_{x=0}$$

$$\xi = x - v^i{}_p t$$
$$\zeta = x + v^r{}_p t$$
$$\eta = x - v^t{}_p t$$

$$f_i(-v^i{}_p t) + f_r\,(v^r{}_p t) = f_t\,(-v^t{}_p t)$$

$$-f_i\big(-v^i{}_p t\big)dt + f_r\big(v^r{}_p t\big)dt = -f_t\big(-v^t{}_p t\big)dt$$

$$-\frac{f_i\big(-v^i{}_p t\big)}{v^i{}_p}d(-v^i{}_p t) + \frac{f_r\big(v^r{}_p t\big)}{v^r{}_p}d(v^r{}_p t) = -\frac{f_t\big(-v^t{}_p t\big)}{v^t{}_p}d(-v^t{}_p t)$$

3.8 REFLECTION AND TRANSMISSION

$$\int -\frac{f_i}{{v^i}_p} d(-{v^i}_p t) + \frac{f_r}{{v^r}_p} d({v^r}_p t) = \int -\frac{f_t}{{v^t}_p} d(-{v^t}_p t)$$

$$-\frac{1}{{v^i}_p} u_I(-{v^i}_p t) + \frac{1}{{v^r}_p} u_R({v^r}_p t) = -\frac{1}{{v^t}_p} u_T(-{v^t}_p t)$$

$${v^i}_p = {v^r}_p = \sqrt{\frac{T}{\rho^i}}$$

$$-\frac{1}{{v^i}_p} u_I(-{v^i}_p t) + \frac{1}{{v^r}_p} u_R({v^r}_p t) = -\frac{1}{{v^t}_p} u_T(-{v^t}_p t)$$

3.8 REFLECTION AND TRANSMISSION

$$-v^t{}_p u_I\left(-v^i{}_p t\right) + v^t{}_p u_R\left(v^r{}_p t\right) = -v^i{}_p u_T\left(-v^t{}_p t\right)$$

$$u_I(0+,t) + u_R(0+,t) = u_T(0-,t)$$

$$u_I\left(-v^i{}_p t\right) + u_R\left(v^r{}_p t\right) = u_T\left(-v^t{}_p t\right)$$

$\times\, v^t{}_p$

$\times\, v^i{}_p$

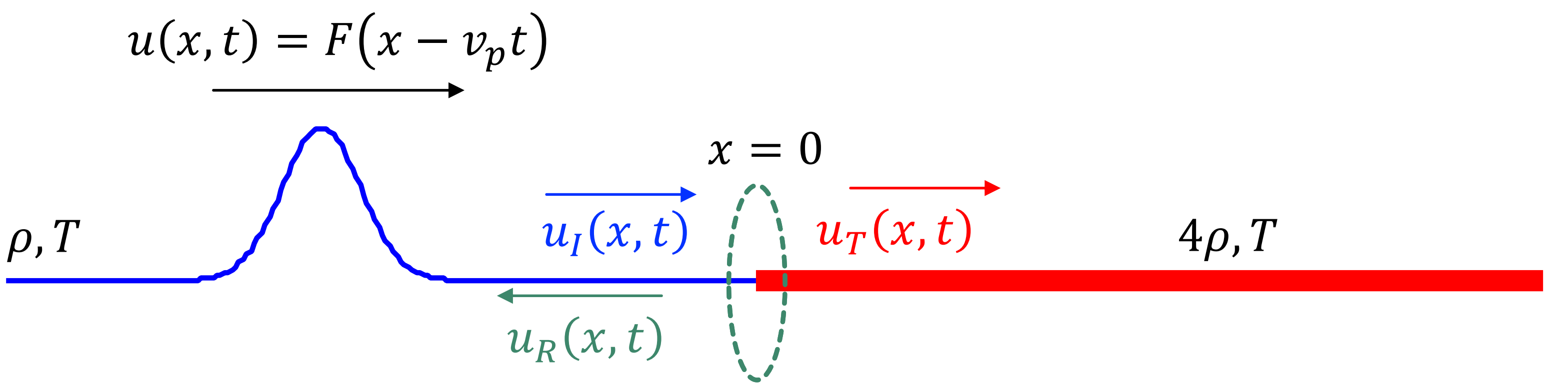
$$2v^t{}_p u_I = (v^t{}_p + v^i{}_p) u_T$$

$$\frac{u_T}{u_I} = \frac{2v^t{}_p}{v^i{}_p + v^t{}_p}$$

$$(v^i{}_p - v^t{}_p)\, v^t{}_p u_I + \left(v^t{}_p + v^i{}_p\right) u_R = 0$$

$$\frac{u_R}{u_I} = \frac{v^t{}_p - v^i{}_p}{v^i{}_p + v^t{}_p}$$

3.8 REFLECTION AND TRANSMISSION



$$R = \frac{u_R}{u_I} = \frac{v^t{}_p - v^i{}_p}{v^i{}_p + v^t{}_p}$$

$$v^i{}_p = v^r{}_p = 2v^t{}_p$$

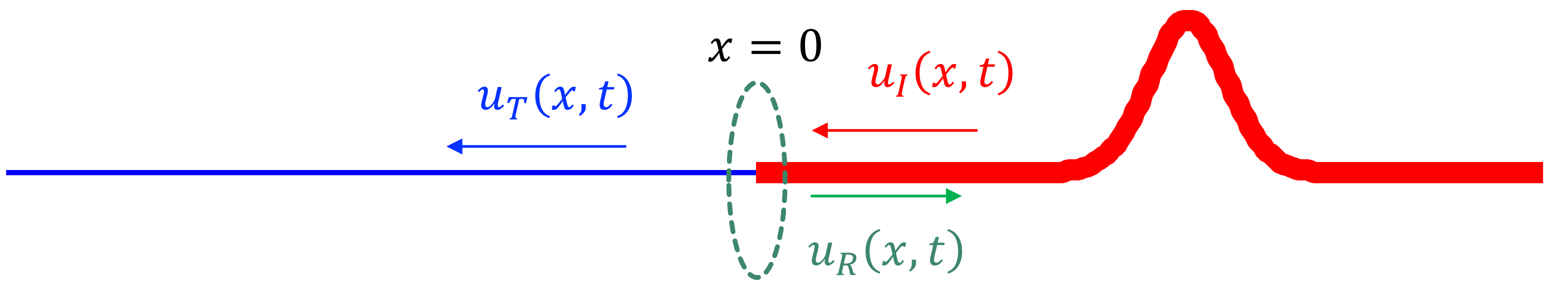
$$R = -\frac{1}{3}$$

$$T = \frac{u_T}{u_I} = \frac{2v^t{}_p}{v^i{}_p + v^t{}_p}$$

$$v_p = \sqrt{\frac{T}{\rho}}$$

$$T = \frac{2}{3}$$

3.8 REFLECTION AND TRANSMISSION



$$R = \frac{u_R}{u_I} = \frac{v^t{}_p - v^i{}_p}{v^i{}_p + v^t{}_p}$$

$$v^i{}_p = v^r{}_p = \frac{1}{2} v^t{}_p$$

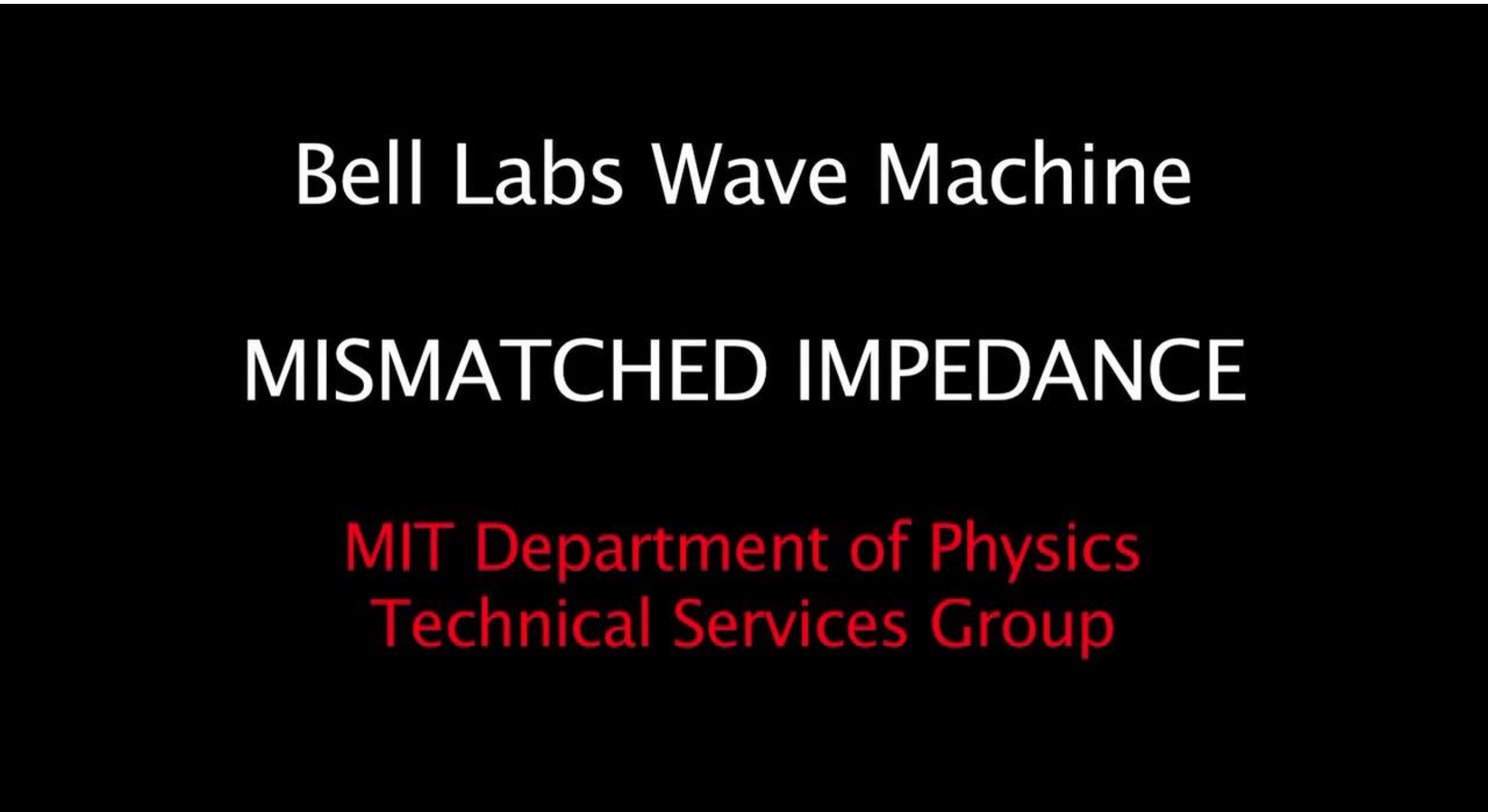
$$R = \frac{1}{3}$$

$$T = \frac{u_T}{u_I} = \frac{2v^t{}_p}{v^i{}_p + v^t{}_p}$$

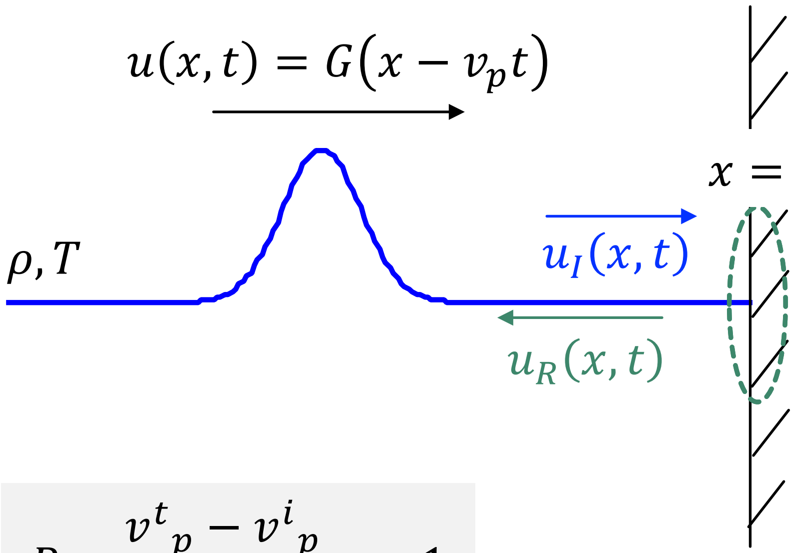
$$v_p = \sqrt{\frac{T}{\rho}}$$

$$T = \frac{4}{3}$$

3.8 REFLECTION AND TRANSMISSION



3.8 REFLECTION AND TRANSMISSION



$$\rho = \infty$$

$$v_p = \sqrt{\frac{T}{\rho}}$$

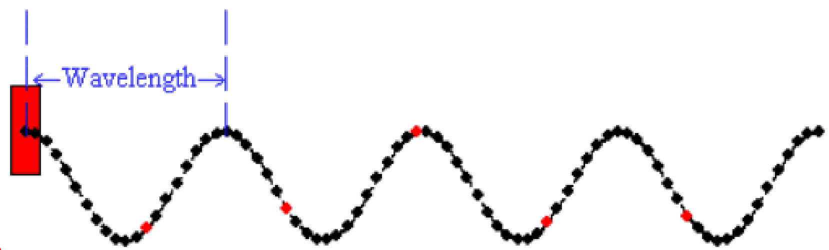
$$v^t{}_p = 0$$

$$R = \frac{v^t{}_p - v^i{}_p}{v^i{}_p + v^t{}_p} = -1$$

$$T = \frac{2v^t{}_p}{v^i{}_p + v^t{}_p} = 0$$

3.9 SOUND AND LONGITUDINAL WAVE

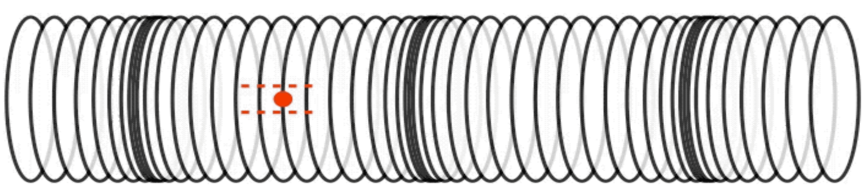
Transverse Wave



displacement

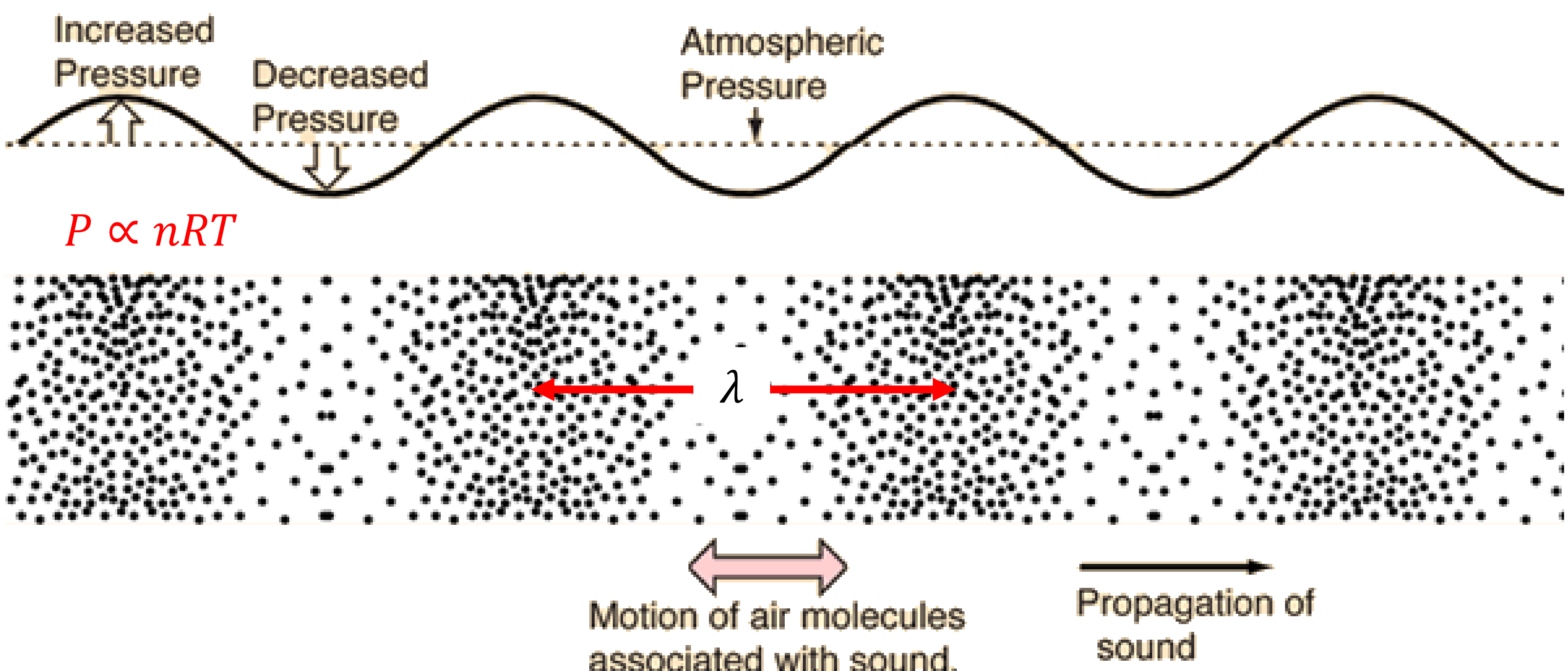
Propagation velocity

Longitudinal wave



3.9 SOUND AND LONGITUDINAL WAVE

Single-frequency sound wave



$$\frac{\omega}{k} = v_p \qquad \lambda = \frac{2\pi}{k}$$

http://hyperphysics.phy-astr.gsu.edu/hbase/Sound/tralon.html

3.9 SOUND AND LONGITUDINAL WAVE

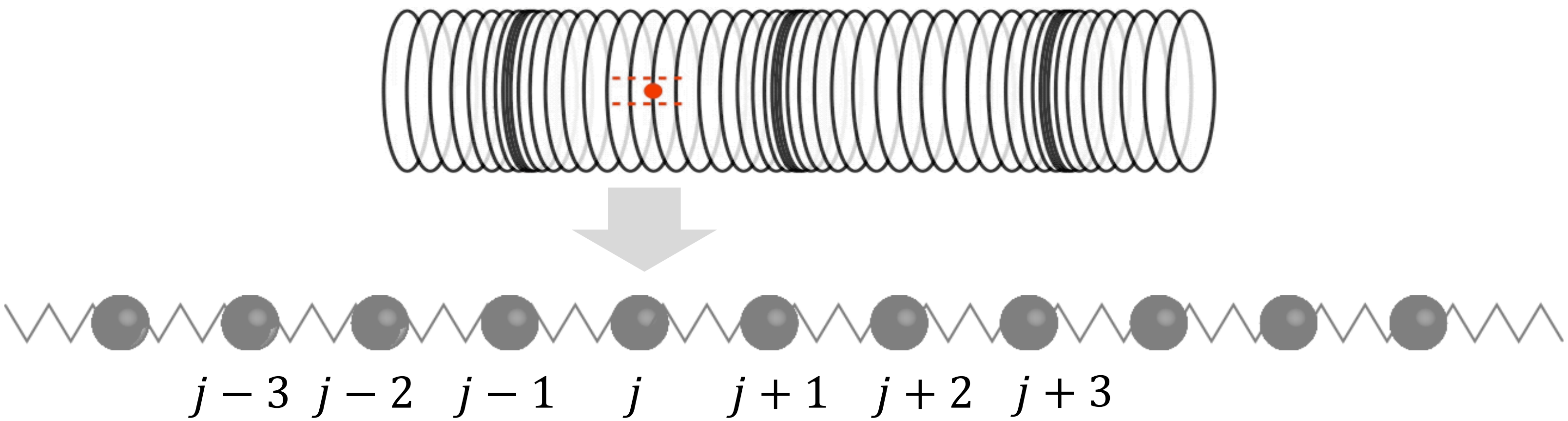
One demonstration that sound is longitudinal

Which one has a higher frequency?

$$\frac{\omega}{k} = v_p \qquad \lambda = \frac{2\pi}{k}$$

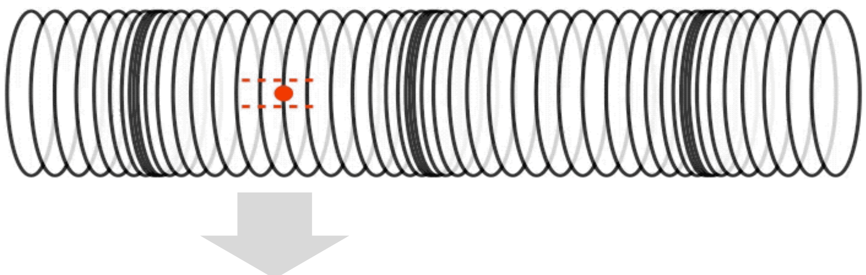
http://hyperphysics.phy-astr.gsu.edu/hbase/Sound/tralon.html

3.9 SOUND AND LONGITUDINAL WAVE



$$m\ddot{u}_j = -K_a\left(u_j - u_{j-1}\right) - K_a\left(u_j - u_{j+1}\right)$$

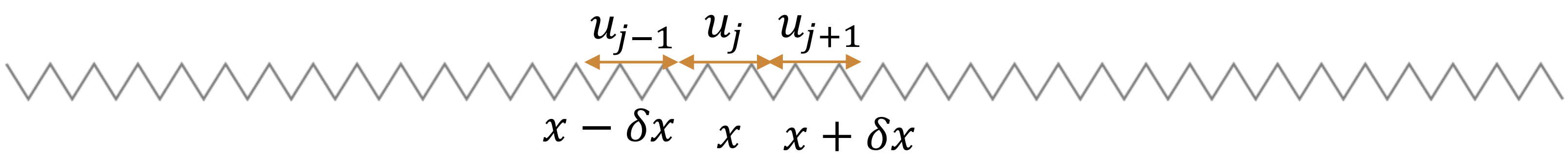
3.9 SOUND AND LONGITUDINAL WAVE



$j-3$ $j-2$ $j-1$ j $j+1$ $j+2$ $j+3$

$$m\ddot{u}_j = -K_a\left(u_j - u_{j-1}\right) - K_a\left(u_j - u_{j+1}\right)$$

3.9 SOUND AND LONGITUDINAL WAVE



$$m\ddot{u}_j = -K_a(u_j - u_{j-1}) + K_a(u_{j+1} - u_j) \qquad m = \rho \cdot \delta x \qquad K_a = K_l \frac{l}{\delta x}$$

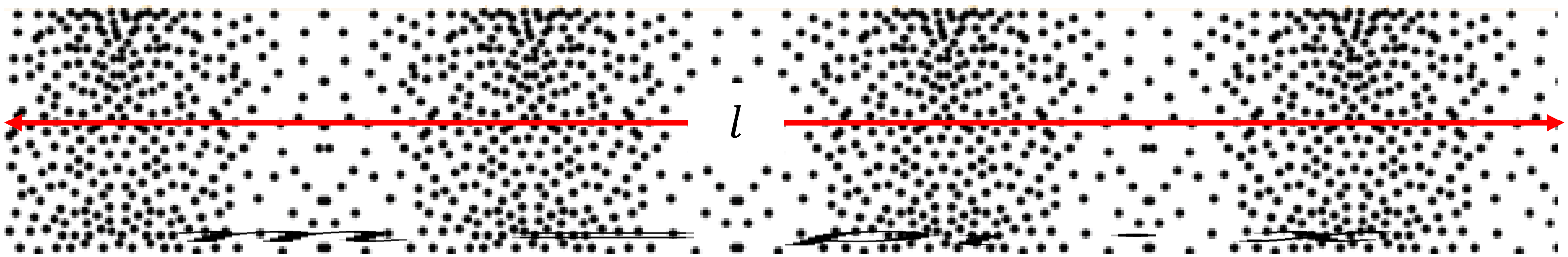
$$\rho \cdot \delta x \cdot \ddot{u}_j = -K_l \frac{l}{\delta x} \delta u_j + K_l \frac{l}{\delta x} \delta u_{j+1}$$

$$\ddot{u}\Big|_x = \left[-K_l l \frac{\delta u}{\delta x}\Big|_{x=x} + K_l l \frac{\delta u}{\delta x}\Big|_{x=x+\delta x} \right] / (\rho \cdot \delta x) = \frac{K_l l}{\rho} \left(u'\Big|_{x+\delta x} - u'\Big|_x \right) / \delta x = \frac{K_l l}{\rho} u''\Big|_x$$

$$\frac{\partial^2 u}{\partial t^2} = \frac{K_l l}{\rho} \frac{\partial^2 u}{\partial x^2}$$

Longitudinal wave equation

3.9 SOUND AND LONGITUDINAL WAVE



$$\frac{\partial^2 u}{\partial t^2} = \frac{K_l l}{\rho_l} \frac{\partial^2 u}{\partial x^2}$$

$$m = \rho_0 \cdot V \qquad m = \rho_l \cdot x \qquad \rho_l = \rho_0 \cdot A$$

$$dF = -K_l dl$$

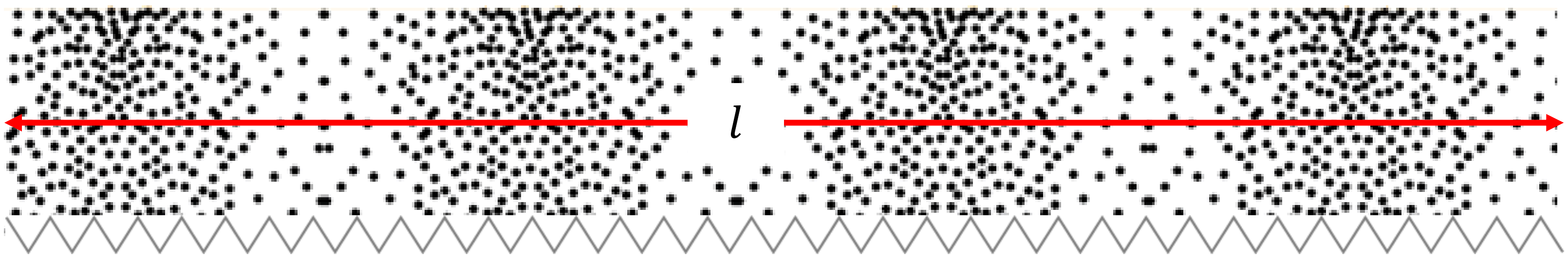
$$\frac{K_l l}{\rho_l} = -\frac{l}{\rho_0 \cdot A} \frac{dF}{dl} = -\frac{l}{\rho_0 \cdot A} \frac{AdP}{dl} = -\frac{1}{\rho_0} V \frac{dP}{dV} = \frac{K}{\rho_0}$$

$$dP = -K \frac{dV}{V}$$

K is the bulk modulus

$$\frac{\partial^2 u}{\partial t^2} = \frac{K}{\rho_0} \frac{\partial^2 u}{\partial x^2} \qquad v_p = \sqrt{\frac{K}{\rho}}$$

3.9 SOUND AND LONGITUDINAL WAVE



Ideal gas: isothermal $PV = B = nRT$

Sound travels fast **Adiabatic** $PV^{\gamma} = C$

$$V^{\gamma} dP + \gamma P V^{\gamma - 1} dV = 0$$

$$dP = -\gamma P \frac{dV}{V}$$

$$K = \gamma P$$

$$v_p = \sqrt{\gamma \frac{P}{\rho_0}}$$

3.9 SOUND AND LONGITUDINAL WAVE

$$v_p = \sqrt{\gamma \frac{P}{\rho_0}}$$

adiabatic index: $\gamma = \frac{C_p}{C_v} = \frac{f+2}{f}$

Monoatomic: $f = 3$

Diatomic: $f = 3 + 2 + 1 = 5 + 1$

The vibrational degree of freedom is not involved, except at high temperatures

air: $\gamma \approx \frac{5+2}{5} = 1.4$

Atmospheric pressure : $P = 1.01 \times 10^5 \text{ Pa}$

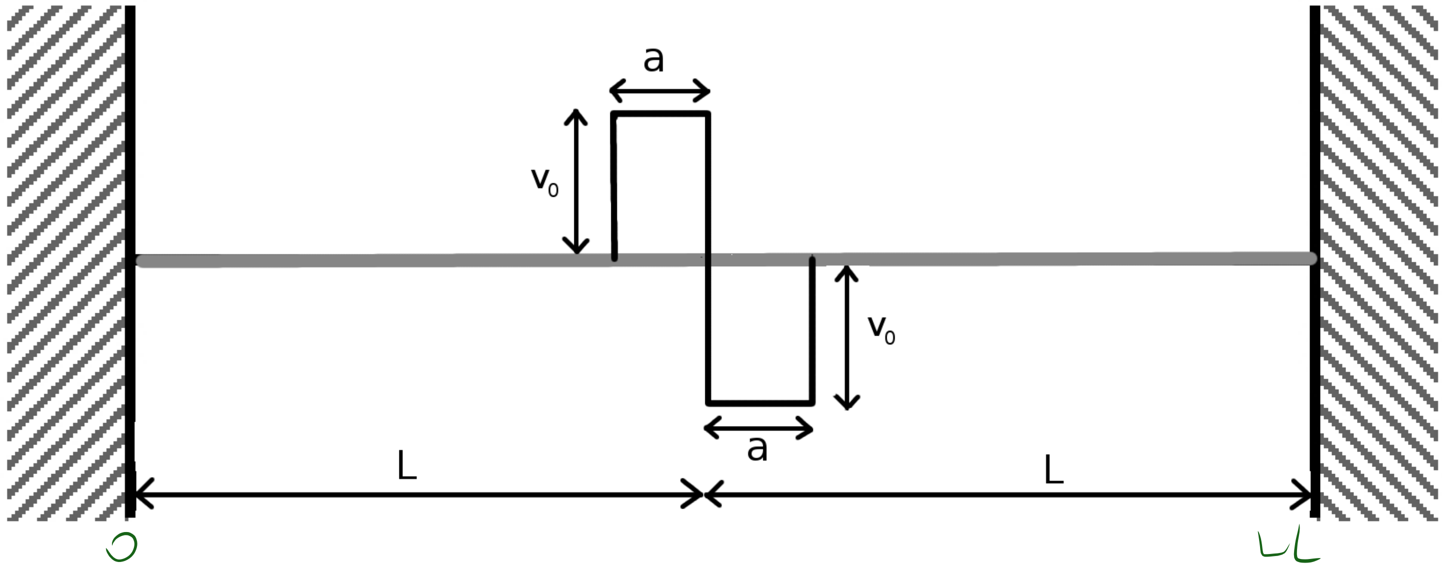
Air density: $\rho_0 = 1.2\ kg/m^3$

Isothermal: $v_p = \sqrt{1 \cdot \frac{P}{\rho_0}} = 290\ m/S$

Adiabatic $v_p = \sqrt{1.4 \cdot \frac{P}{\rho_0}} = 343\ m/S$

According to Laplace, when sound travels in a gas, there is no time for heat conduction in the medium, and so the propagation of sound is adiabatic.

HOMEWORK



$$y(x, t=0) = 0$$

$$\dot{y}(x, t=0) = \begin{cases} v_0 & : \quad L-a \le x < L \\ -v_0 & : \quad L \le x < L+a \\ 0 & : \quad \text{elsewhere} \end{cases}$$

通解：

$$y(x,t) = \sum_{n=1}^{\infty} A_n \sin(k_n x + \varphi) \sin(\omega_n t + \phi)$$

Amplitude of nth normal mode? Lowest unexcited mode?

$$y(x,0) = \sum_{n=1}^{\infty} A_n \sin(k_n x + \varphi) \sin(\phi) = 0 \qquad \phi = 0$$

$$y(x,t) = \sum_{n=1}^{\infty} A_n \sin(k_n x + \varphi) \sin(w_n t)$$

而 $y(0,0) = y(2L,0) = 0$

$$\therefore y(x,t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi}{L} x\right) \sin(w_n t)$$

$$A_n = \frac{2}{2a} \int_{L-a}^{L+a} y(x,t) \sin\left(\frac{2\pi}{2a} nx\right) dx$$

$$= \frac{1}{a} \int_{L-a}^{L} v_0 x \sin\left(\frac{n\pi x}{a}\right) dx + \frac{1}{a} \int_{L}^{L+a} -v_0 x \sin\left(\frac{n\pi x}{a}\right) dx$$

$$y(x,t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) \cos(\omega_n t + \varphi_n)$$

$$y(x,0) = 0 \qquad \dot{y}(x,0) = 0 \quad (x < L-a \text{ or } x > L+a)$$

$$\therefore \quad A_n = \frac{2}{L}\int_0^L y(x,0) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{2}{L}\int_{L-a}^{L} V_0 \sin\left(\frac{n\pi x}{L}\right) dx + \frac{2}{L}\int_{L}^{L+a} (-V_0) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{2V_0}{L}\left[-\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\right]_{L-a}^{L} + \frac{2V_0}{L}\left[\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\right]_{L}^{L+a}$$

$$= \frac{2V_0}{n\pi}\left(\cos\left(\frac{n\pi(L-a)}{L}\right) - \cos\left(\frac{n\pi(L+a)}{L}\right)\right) = \frac{4V_0}{n\pi} \sin n\pi \sin \frac{n\pi a}{L}$$

lowest unexcited $n=1$: $\quad A_1 = \frac{2V_0}{\pi}\left(\cos\left(\frac{\pi(L-a)}{L}\right) - \cos\left(\frac{\pi(L+a)}{L}\right)\right)$

$$= 0$$